ECEN 520 Midterm Project Report: Analyzing Wearable Sensor Data for Occupational Risk Analysis

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Abstract

Work-related musculoskeletal injuries are a growing concern in physically demanding occupations, particularly those involving repetitive or excessive lifting. This project analyzes wearable electromyography (EMG) sensors to assess biomechanical strain during lifting tasks. Two subjects participated in controlled lifting experiments involving low-risk (light) and high-risk (heavy) box lifts, with EMG sensors placed on both left and right biceps, and both left and right deltoids. From the EMG signals, we extracted statistical features such as mean, standard deviation, min, max, and RMS for each lifting trial. These features were used to train three supervised machine learning models: Logistic Regression (LR), Decision Tree (DT), and K-Nearest Neighbors (KNN). These models were used to classify each lift as high-risk or low-risk.

KNN achieved the highest accuracy and F1 scores, outperforming the other two models. The results ultimately suggest our KNN model was the best for classification of EMG data for high-risk vs low-risk lifting tasks. More research should be done in the future regarding other types of models such as CNNs and with larger datasets than what we used. [1]

[1]This project was completed as part of the midterm project for ECEN 520: Introduction to Machine Learning, supervised by Prof. Fatemeh Davoudi. The data used in this project was collected using equipment provided by Prof. Davoudi's lab, is confidential and is intended solely for this course. Students are not permitted to share or distribute the data for any other purposes.

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1 Introduction

In this project, we looked into how electromyography (EMG) might be used to gauge the danger of physical tasks by measuring muscle activity. Participants completed a repetitive lifting activity in two different conditions, one "high-risk" and one "low-risk", with EMG sensors attached to their bicep and deltoid muscles. The objective was to use machine learning techniques to categorize the muscle activity data according to each task type, and determine which model was the best fit for our data and experiment. In order to investigate the viability of employing EMG signals and machine learning models for ergonomic risk assessment and injury prevention, we used K-Nearest Neighbors (KNN), Logistic Regression, and Decision Tree classification models to differentiate between high-risk and low-risk movement patterns. Initially, our hypothesis was that the Logistic Regression model would be the most effective classifier since we required only a binary classification.

2 Data Understanding and Preprocessing

2.1 Dataset Description

The data for this study was collected in the Santa Clara Machine Learning and Safety Analytics Lab on May 8, 2025 and involved participants performing a repetitive lifting exercise in which they moved a large box from waist-height to head-height. The full data set includes two male participants lifting 5kg and 10kg weights for 20 repetitions each, totaling 80 instances for the whole study. These repetitions were timed to 4-second intervals and their biometrics were recorded. This resulted in a total of four time-series data files:

- P1_High_2_0.csv — Participant 1 High Risk Lifting (10kg)
- P1_Low_2_0.csv — Participant 1 Low Risk Lifting (5kg)
- P2_High_2_0.csv — Participant 2 High Risk Lifting (10kg)
- P2_Low_2_0.csv — Participant 2 Low Risk Lifting (5kg)

The participants biometrics were sampled at 1259 Hz to capture high-frequency muscle activation dynamics in the upper body at the following locations:

- Sensor 1 — Right Bicep
- Sensor 2 — Right Deltoid
- Sensor 3 — Left Bicep
- Sensor 4 — Left Deltoid

Gyroscope and Accelerometer data was also collected from these sensors, but due to our test conditions and chosen training algorithms, this data was not used for the folloiwng analysis.

Each CSV file includes raw time series data for each of the EMG sensors, as shown in Figure 1. Each sensor has a value column representing the activation strength at that point in time, and a timestamp column corresponding to when the data point was taken.

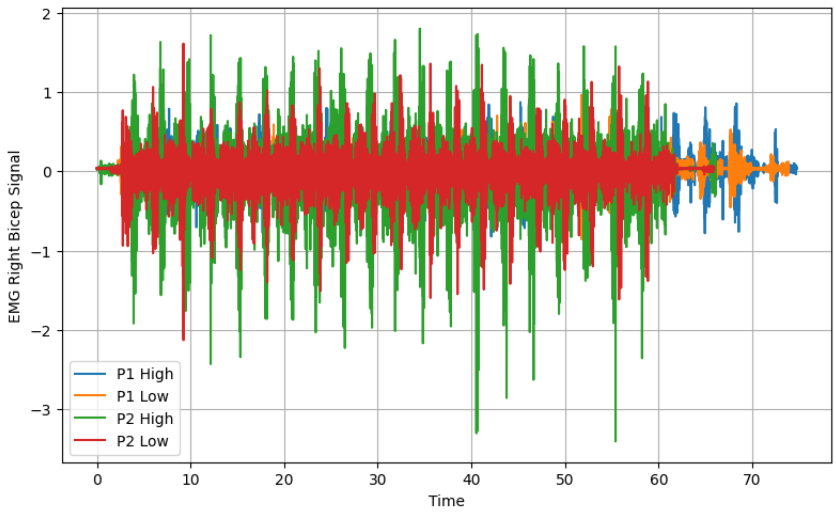

Figure 1: Time series data collected from the EMG sensor placed on the participants' right biceps. This contains signals from the 4 trials.

2.2 Data Cleaning

The data was cleaned carefully in the following ways:

- Unused data removed
- Redundant columns removed
- Leading and trailing space removed
- Signals split into single-lift intervals
- Features extracted

As the gyroscope and accelerometer data was not used for this training, these columns were dropped, as were the identical timestamp columns on each EMG sensor. The data was determined to have been trimmed a the beginning and end of each trial properly, so there was little excess data to remove. The data was iteratively split into 4-second intervals which were stored in their own dataframes, and finally, the following features were extracted.

2.3 Feature Engineering

The features in Tabel 1 below were extracted from each 4-second interval and used as the case iterations for the algorithm training. Of particular note in this high-risk repetitive task study are max, standard deviation, and mean. When assessing the risk factor of a task, the maximum muscle strain is hypothesized to be a primary indicator. Similarly, a high root mean squared would indicate extended periods of muscle activation, and a high standard deviation would correspond to regular switches between high and low muscle activation, and a high variation between the two.

The final 80 instances were split 70/30 between training data and testing data respectively with the same random state being used for all three models to ensure equality in data splitting.

Table 1: Sample Feature Extraction. These data points were collected by the EMG sensor on participant 1's left bicep during the 7th repetition of the low-risk lift and the corresponding repetition of the high-risk lift. As predicted, the maximum, standard deviation, and root mean squared predictors look promisingly different for the two risk levels.

Trial	**Min**	**Max**	**Mean**	**St. Dev**	**RMS**
Low Risk	-1.427	0.861	-0.003	0.131	0.131
High Risk	-2.182	1.585	-0.003	0.287	0.287

3 Modeling

3.1 Model Selection

For our project, we were allowed to select from a set of five different classification models to work on our data. Those algorithms include Logistic Regression, Naive Bayes, K-Nearest Neighbors (KNN), Decision Trees, and Random Forest. We chose to neglect Naive Bayes and Random Forest and focus only on the other three. We ignored Naive Bayes because the model assumes that all features contributing to the classification are independent of each other. This does not fit our data since the bicep and deltoid muscles are physically connected, meaning strain on one muscle will inevitably affect the other muscle, meaning they are not independent features. We chose to ignore Random Forest classification because it trains multiple decision trees on random subsets of our data. Since we have a low number of data points, we believe that it is more effective to sample a single decision tree on the entire dataset instead of using Random Forest Classification.

3.2 Model Training

3.2.1 Logistic Regression

The logistic regression training was conducted using ridge regression. The values in Table 2 correspond to the weights generated by the logistic regression model. Unsurprisingly, the mean features had low coefficients. The magnitude signals are symmetric making the mean of all signals roughly zero. These features therefore had little effect on the classification. Also as a result of this symmetry, the min and max features had comparable coefficient representing strong collinearity. The min and mean features were therefore removed without consequence for the output.

3.2.2 Decision Tree

The decision tree algorithm was trained using a hard-coded random state and stratified y values. The minimum required samples to split was set to 15 because it produced a much simpler tree (shown in Figure 2) with no cost to the accuracy of the model.

3.2.3 K-Nearest Neighbors

The K-nearest neighbors training had all parameters set to the default values. These defaults are set by scikit-learn because they tend to yield the best results, especially on small to medium sized datasets like ours [4].

Table 2: Logistic Regression Weights.

Term	**Coefficient**
Intercept	5.8625
Left Delt Mean	-1.6576e-04
Left Delt Max	-4.7428e-01
Left Delt Min	3.3252e-01
Left Delt Std Dev	-1.1818e-02
Left Delt RMS	-1.0380e-02
Right Delt Mean	-4.4925e-04
Right Delt Max	-1.7025e-01
Right Delt Min	3.3332e-01
Right Delt Std Dev	-1.0810e-02
Right Delt RMS	-1.0767e-02
Left Bicep Mean	1.6552e-03
Left Bicep Max	1.7288e-01
Left Bicep Min	5.3909e-01
Left Bicep Std Dev	-9.6480e-01
Left Bicep RMS	-9.6474e-01
Right Bicep Mean	5.2871e-04
Right Bicep Max	-1.1958e+00
Right Bicep Min	1.6373e+00
Right Bicep Std Dev	-2.5025e-01
Right Bicep RMS	-2.3184e-01

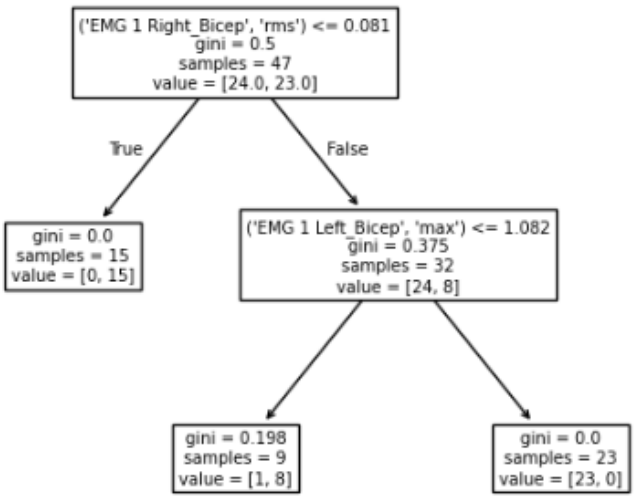

Figure 2: Decision Tree Model

3.3 Evaluation Metrics

For every model, we judged its effectiveness at classification with the F1-score metric. F1-score is close to 1 when the number of true positives is much greater than the number

of false positives and false negatives, and close to 0 when the opposite is true. We want our classification algorithms to have a low number of false positives and false negatives, so a F1-score of as close to 1 as possible is desired.

4 Results and Discussion

4.1 Model Performances

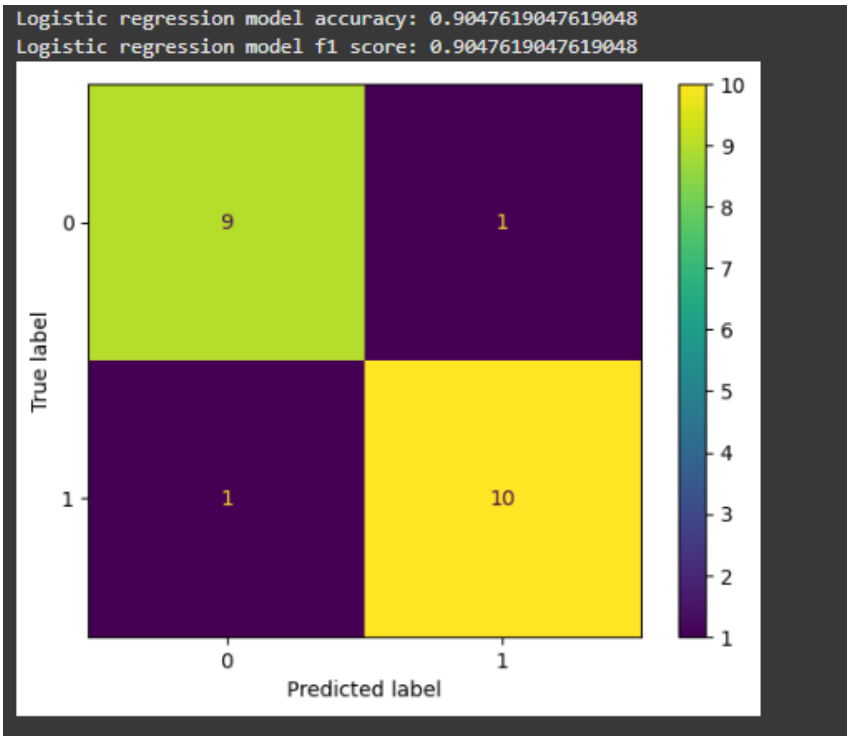

Figure 3: LR Model Performance

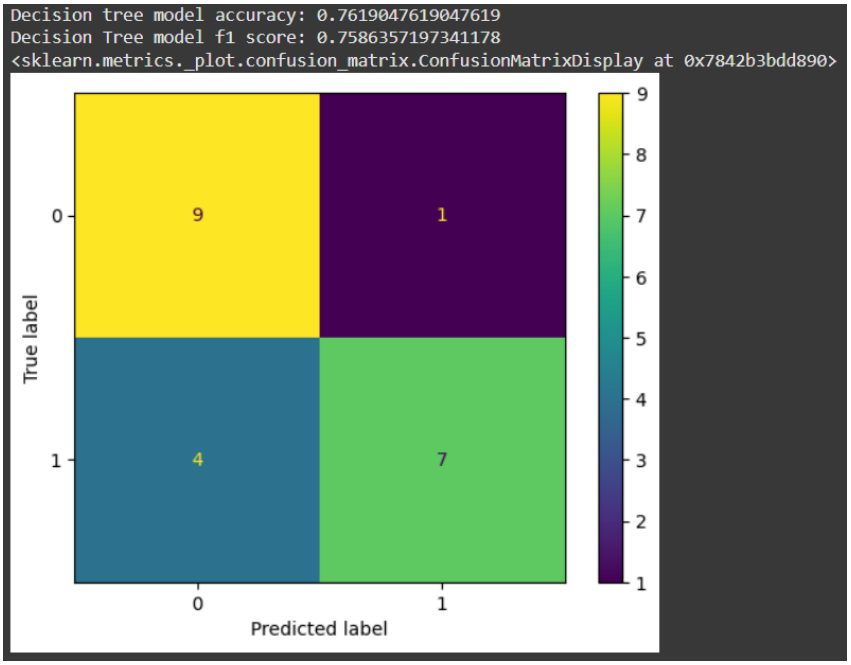

Figure 4: DT Model Performance

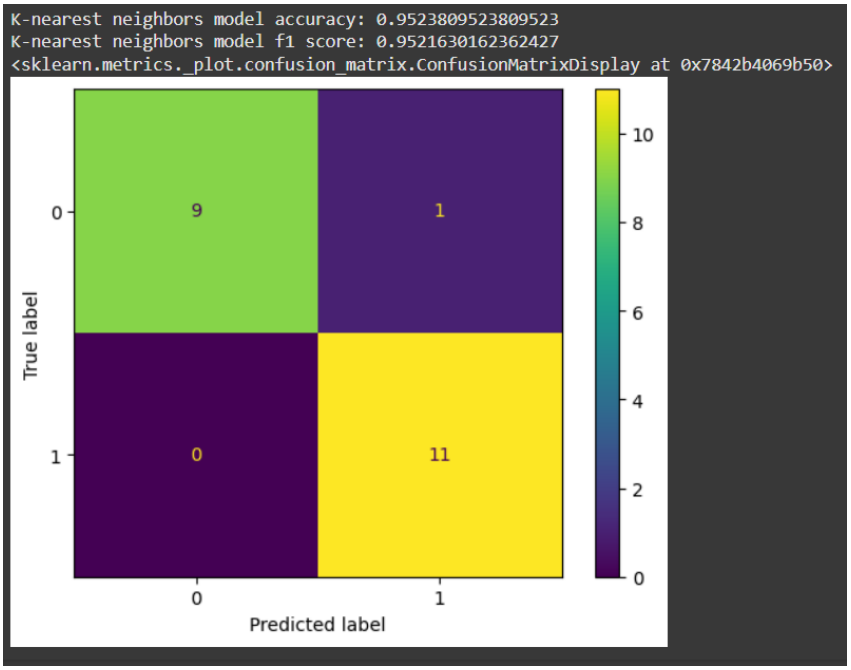

Figure 5: KNN Model Performance

4.2 Discussion

The K-Nearest Neighbors (KNN) model achieved accuracy and f1 scores of 95.2%, which outperformed both the Logistic Regression (LR) and Decision Tree (DT) models. We only see one misclassification on the confusion matrix for KNN as well. From this, we can conclude that KNN was effectively able to classify the nonlinear relationships in EMG data. This makes sense given the nature of the problem. EMG signals are not linearly separable, but they don't need to be for KNN as it classifies based on local patterns in feature space, which fits our well separated dataset.

Initially, the LR model achieved accuracy and f1 scores of 76.2% as it misclassified 5 samples. This lower score seemed to show how LR struggles more when given non-linear data. However, after re-running the model using Ridge Regularization, the Logistic Regression model significantly improved and achieved 90% accuracy, reducing the gap with KNN. This suggests that regularization helped the model generalize better and reduced the effect of multicollinearity in our features.

Similarly, the DT model achieved an accuracy score of 76.2%, but the f1 score was slightly lower at 75.9%. We were quite surprised by this low accuracy as DT models are often used to model nonlinear relationships. After some research, we realized the dataset was likely too small and that this led to overfitting [2]. The model was overfitting some patterns while not generalizing others, and that's what led to it misclassifying 5 different cases. If we had a larger dataset, it's very likely the DT model would've performed better.

5 Conclusion

Ultimately, we conclude that KNN is the most effective model among our three tested to determine high-risk vs low-risk lifting actions from EMG sensor data. It performed the best for all of our metrics with no observable performance tradeoff. If we were to do this experiment again, we would test out a Convolutional Neural Network (CNN) model. A CNN would hypothetically apply filters over these EMG signals and learn the spatial/temporal patterns needed for classification. They are also often used in existing EMG classification systems [1]. We would also try to increase our dataset so the DT model wouldn't have an overfitting problem.

6 References

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